Average rates, derivation and integration for functions with one independent variable

Author: Cecilia Alvares

1) Average rate function

For whichever function $f(x)$, one can assemble (i.e., define) a second function to quantify the average rate of the variation of y within a specific given interval of variation of x based on the initial and final values of y within that interval. The definition of average rate is as follows:

Given an interval of variation in x defined by $\Delta x = x_F - x_I$, the value of average rate is the variation observed in y within the scope of $f(x)$ per unit of x such that if it was kept constant throughout the entire corresponding interval length Δx, the proper corresponding variation in y, given by $\Delta y = y_F - y_I = f(x_F) - f(x_I)$, would be obtained.

We refer to the function that gives the average rate of $f(x)$ as the average rate function. From understanding the definition above and the different mathematical operations we have defined, it is not difficult to see that the expression that fundamentally defines the **average rate function**, symbolized as $m(x, \Delta x)$, for any general function $f(x)$ is given by equation (1):

$$m(x, \Delta x) = \frac{f(x + \Delta x) - f(x)}{\Delta x} \tag{1}$$

The average rate function allows the variation $f(x)$ at a given desired interval of the domain, $(x, x + \Delta x)$, to be obtained by a multiplication shown in equation (2), thus showing that equation (1) is indeed reflecting the definition of an average rate function.

$$\Delta y = m(x, \Delta x)\Delta x \tag{2}$$

The function $m(x, \Delta x)$ as defined in equation (1) depends on the domain interval length, Δx, and on the initial value of x used to define it: this is an alternative to defining the average rate as a function of x_I and x_F instead. We note that $m(x, \Delta x)$ can be used to evaluate average rates for a given punctual interval but its values for different combinations $(x, \Delta x)$ can used sequentially to evaluate how $f(x)$ changes in a series of consecutive intervals marked each by a given $(x, \Delta x)$. In the latter case, it is necessary to ensure that the values possible for $(x, \Delta x)$ are indeed such that they yield consecutive intervals of values in the domain, meaning x and Δx no longer vary independently of each other.

Practical example: imagine that we managed to write a function for how the distance ran by a car changes with time, $d = f(t)$ (the specific expression of the function is not important here). We may want to know the average speed with which the car moves from the start to the end of the trip. This could be accomplished by splitting the total time taken to complete the trip in intervals of length Δt, which may or may not have a uniform length, computing the distance Δd ran in each interval and calculating the ratio $\Delta d/\Delta t$ for each interval.

It is perhaps interesting to note the fact that the algebraic pattern of the function $f(x)$ is tied to the algebraic pattern of the function $m(x, \Delta x)$, given by combinations of input and output $(x, \Delta x, \Delta y)$, as these two functions are linked. In fact, in a way, the pattern of $m(x, \Delta x)$ can be regarded as a different way to portray the pattern of $f(x)$: whilst the function $f(x)$ itself maps the correspondence between x and y, given the expressions that appear on its numerator and denominator, the average rate function allows to more straightforwardly understand how deeply the variation of x affects the variation of y as well as the regions in which the impact is higher or lower.

2) Derivative function

Going deeper into the idea of defining average rate functions, one could start dwelling on the idea of an infinitesimal (or instantaneous, if the IV is time) average rate function, namely the one that describes the average rate for interval lengths $\Delta x = 0$. **In principle, the truth is that any function could be any. This is so because in an interval $\Delta x = 0$, the change in value of any function $f(x)$ is also 0, meaning that any value of rate, if kept constant for $\Delta x = 0$, will satisfy a total variation of 0 in $f(x)$.** Indeed, within equation (1), this scenario would lead to 0/0 which is a well know indeterminate form by now: as discussed in a previous document, any value can be an output for such operation.

Despite the technical fundamental infinite amount of representatives for an infinitesimal average rate function of $f(x)$, **a convention** was made to define it as the limit of function $m(x, \Delta x)$ as $\Delta x \to 0$. The infinitesimal average rate function conventionalized as referred to as **derivative**, and its definition as well as many symbolisms for it, is shown in equation (3):

$$f'(x) = \frac{dy}{dx} = \frac{df}{dx} = \lim_{\Delta x \to 0} m(x, \Delta x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \tag{3}$$

Given that $m(x, \Delta x)$ is not a defined 1-IV function for $\Delta x = 0$, it is necessary to counter the indetermination by seeking for a semi-equivalent function" at each value of x, as discussed in the previous chapter. Naturally, the expression, which will be a function of x, should give the value of the limit of $f(x)$ at the given value of x within the domain.

For the most various general function forms, $f(x)$, such semi-equivalent functions and thus the limit in equation (3) was calculated. Such a heavy work allowed to reach a correspondence between the form of whatever given function (e.g., polynomial, trigonometric, logarithmic, exponential) and the form of its derivative. From observing the correspondence, a way of how to determine $f'(x)$ from the form of $f(x)$ was acknowledged, and this gives rise to what are called **derivation rules**. These rules are basically an automatized method for finding the derivative of a function $f(x)$, often referred to as **primitive** of $f'(x)$, without going through the arduous and time consuming task of actually estimating the limit shown in equation (3) every single time. They can be found in any textbook of Calculus – here we focus rather on a conceptual understanding and in a storytelling. It is worth noting that the derivation rules are not something to try to make rational logic sense of,[1] as they a priori are simply as a means of acting on a function $f(x)$ of a given form to get its $f'(x)$.

It is worth noting is that the that in principle an infinite amount of function forms can be created by coupling mathematical operations. This means that the heavy work of deriving a function of a specific form and establishing a derivation rules is limited to a finite number of function forms. For functions for which derivation rules are not available in text books, it is necessary to evaluate the limit as defined in equation (3). In this context, it is worth noting that some properties inherent of a limit can extend to some extent the applicability of the derivation rules to more complicated functions. For example, it was shown in the previous document that the sum of the limit of two separate functions $f(x)$ and $g(x)$ as $x \to a$ is equal to the limit of the function $f(x) + g(x)$ as $x \to a$. The same holds for subtraction. This validity of this property ensures that the derivative of any function, $f(x)$, formed by the sum of two others, $f(x)$ and $g(x)$, for which derivation rules exist can be determined by deriving the individual addends and summing the result, as proved below:

$$h(x) = f(x) + g(x) \tag{3}$$

[1] This means you don't need to try to make sense of why the 2 in an exponent of a polynomial "comes down as a pre-factor". I mean, there is no bright, logical explanation of why this is – the explanation lies on it being a pattern observed in observing the form of a polynomial in contrast with its derivative.

$$h'(x) = \lim_{\Delta x \to 0} \frac{h(x+\Delta x) - h(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) + g(x+\Delta x) - f(x) - g(x)}{\Delta x} \quad (4)$$

$$h'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} + \lim_{\Delta x \to 0} \frac{g(x+\Delta x) - g(x)}{\Delta x} \quad (5)$$

Similarly, the additive and product rules for limits allow to conclude that the derivative of a function, $h(x)$, formed by the product of two other functions, $f(x)$ and $g(x)$, is equivalent to $f(x)g'(x) + f'(x)g(x)$, further extending the utility of the derivation rules provided for any function $h(x)$ formed by $f(x)$ and $g(x)$ for which derivation rules exist. This is shown in equations (6)-(11), in which we note particularly that in equation (10) the function $f(x)$ can be taken out of the limit as it does not depend on Δx, functioning thus as a constant for each value of x.

$$h(x) = f(x)g(x) \quad (6)$$

$$h'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)g(x+\Delta x) - f(x)g(x)}{\Delta x} \quad (7)$$

$$h'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)g(x+\Delta x) - f(x)g(x+\Delta x) + f(x)g(x+\Delta x) - f(x)g(x)}{\Delta x} \quad (8)$$

$$h'(x) = \lim_{\Delta x \to 0} \frac{g(x+\Delta x)\big(f(x+\Delta x) - f(x)\big) + f(x)\big(g(x+\Delta x) - g(x)\big)}{\Delta x} \quad (9)$$

$$h'(x) = \lim_{\Delta x \to 0} g(x+\Delta x) \, . \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} + f(x) . \lim_{\Delta x \to 0} \frac{g(x+\Delta x) - g(x)}{\Delta x} \quad (10)$$

$$h'(x) = g(x)f'(x) + f(x)g'(x) \quad (11)$$

Finally, one thing that may be useful to note within the scope of equation (3) that holds for whatever function form is that functions differing from each other by summation by a constant have precisely the same derivative. This is because this constant appears in the argument of both $f(x + \Delta x)$ and $f(x)$.

Nothing forbids $f(x)$ to be derived multiple times sequentially. Given a function $f(x)$, the function obtained upon deriving its derivative is referred to as second derivative of $f(x)$ and is symbolized as $f''(x)$, $f^{(2)}(x)$ or $d^2f/\,dx^2$. The second derivative is said to be the derivative of order two of $f(x)$. Naturally, the relationship between $f'(x)$ and $f''(x)$ is the same as that of $f(x)$ and $f'(x)$. Higher order derivatives follow the same scheme of symbology as the pattern that can be inferred from the symbology for first and second derivatives. The relationship between whatever pair of functions in the sequence of derivation is given by their derivation order.

3) Symbolizing the variation of the dependent variable in differential form

Just like we can retrieve the variation of y, Δy, within a given interval via $m(x, \Delta x)$, we may also represent the infinitesimal variation of y, dy, using an equation. This gives rise to equation (12). The latter it is a symbolic equation since dx and dy are representative entities and if they had an actual numerical counterpart, it would be zero.

$$dy = \frac{dy}{dx} dx \tag{10}$$

4) Integral of a function

Just like derivation consists of obtaining the derivative of a function, $f'(x)$, as defined in equation (3), from the function $f(x)$, we can also define an operation that allows retrieving the function $f(x)$ from $f'(x)$. This latter operation is called **integration**, and by inverting the previously mentioned derivation rules, one can establish the **integration rules**. In this context, it may be worth noting that since functions differing from each other by a constant addend share the same precise derivative, integration rules cannot fully recover the entire form of the primitive on its own. More information on the original $f(x)$ function would have to be known in order to recover its full format if desired, such as knowledge of a value of $f(x)$ for a given x for example: replacing these in the equation for $f(x)$ would lead to an equation with one unknown, namely the constant. If no additional information is given, the function $f(x)$ obtained through integration only can still be written down, carrying now a $+c$ constant in its expression to indicate that the existence of this term, as well as its value in case it is non-zero, is unknown.

The symbolism for this operation in mathematical language is given below in equation (13). The dx which appears in the notation serves to indicate with respect to which variable the integration operation is being performed – despite the current topic being functions of one IV, having such an identifier becomes important in the context of multi-IV functions, as it will be seen in further chapters.

$$f(x) = \int f'(x) dx \tag{13}$$

Given that integration allows to retrieve the function $f(x)$, it can be used to determine its variation within a given interval of the domain. As this variation is indifferent to the value of a possible constant addend on the expression of $f(x)$, it can be fully determined from integrating $f'(x)$ and calculating $f(x_F) - f(x_I)$. This is often referred to as "integrating $f'(x)$ over an interval" or rather **definite integration**. In this scenario, the limits of this interval, referred to in this case as **limits of integration**, are plugged in the symbology for the integration as pictured in equation (14):

$$f(x_F) - f(x_I) = \int_{x_I}^{x_F} f'(x) dx \tag{14}$$

5) Geometrical aspects of derivation and integration

a) Area underneath $m(x, \Delta x)$, the variation of $f(x)$ and Riemann's sum

As it was previously seen, the function $m(x, \Delta x)$ was algebraically built to have the meaning of average rate of $f(x)$. In other words, when built using a non-zero value of Δx, the corresponding discretized $m(x, \Delta x)$ function gave values that, when multiplied by the given Δx, would return the rightful variation of $f(x)$ within that specific interval marked by initial and final values of x and length Δx. When plotting the average rate function for whichever fixed value of Δx that is different than 0, and naturally with the corresponding discretized values of x that are consecutively spaced by Δx, the graph resembles a bunch of rectangles put together. Take a look at figure 1 for an illustration.

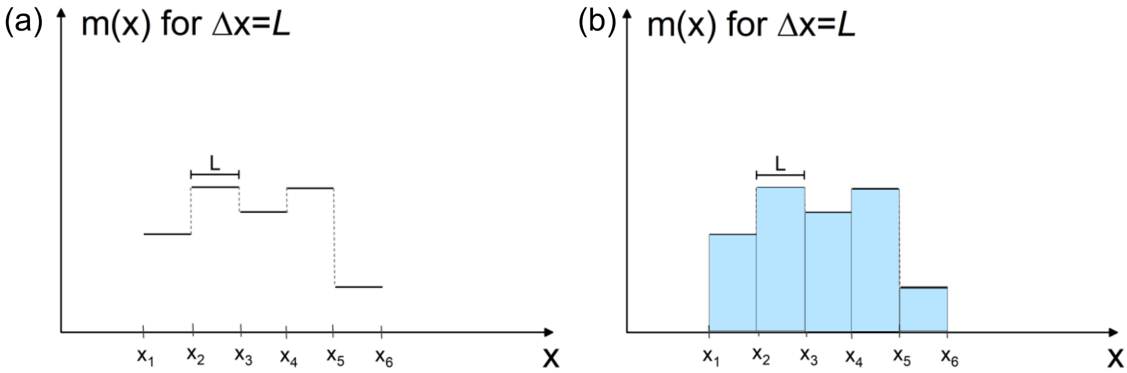

Figure 1. (a) Generic plot of m(x) for $\Delta x = L$ where L is any arbitrary value. It is possible to see that $m(x)$ values are constant for an entire interval of length Δx and then changes its value, as expected from its definition. Panel (b) shows how this plot geometrically resembles a series of adjacent rectangles of length Δx and height equal to $m(x)$ within that x interval.

The area of each and all rectangle of the plot, happens to be exactly the product between $m(x, \Delta x)$ (rectangle height) and Δx (rectangle length). At the same time, this product, $m(x, \Delta x)\Delta x$, corresponds to the variation of the function $f(x)$ whose average rate is quantified by $m(x, \Delta x)$ in the given Δx interval. **This means that despite being parallel notions, the area underneath the graph of $\boldsymbol{m(x, \Delta x)}$ built with $\boldsymbol{\Delta x \neq 0}$ within a given overall interval $\boldsymbol{[x_I, x_F]}$ is equal to the variation of the function $\boldsymbol{f(x)}$ in that same interval, $\boldsymbol{y_F - y_I}$.** The exception to the rule is when $m(x, \Delta x)$ is negative: in this case the area of the rectangle should take into consideration the absolute value of the triangle's height. This exception aside, the property of having the area underneath the graph of $m(x, \Delta x)$ match the variation of the function $f(x)$ holds always, since the expression for the area of the rectangle doesn't change nor does the equivalency between $m(x, \Delta x)\Delta x$ and Δy.

However, the link between the area underneath the $m(x, \Delta x)$ and the variation of $f(x)$ breaks, in principle, at the limit case $\Delta x \to 0$. The rupture happens because the region underneath $f'(x)$ is no longer composed by a set of rectangles put together: the rectangles have no longer a finite length. Therefore, there is no guarantee from basic geometry about that equivalency in a given interval of the domain. To evaluate if the link still holds, it would be necessary to find an alternative expression for the area underneath the $f'(x)$'s curve and confirm it matches the variation in $f(x)$. However, in face of the complex shape of the plot of functions, there is no formula predefined from basic geometry to calculate the area underneath its curve.

To tackle the task, the Riemann sum comes in hand. The Riemann sum is an expression that sums the areas of adjacent rectangles placed underneath a function's plot, with the function value at a given point being a rectangle height and the length being the Δx. In our particular context, the function is going to be the derivative of $f(x)$, $f'(x)$, although the Riemann sum can conceptually be about any function. Figure 2(a) aims to illustrate the previously given definition of the Riemann's sum in the context of $f'(x)$ and the corresponding expression for the Riemann sum is given in the right hand side of equation (15). In equation (15), discrete values of $f'(x)$ (or any other function, for a general Riemann sum) are considered depending on the spacing Δx that is chosen. It is also worth noting that the Riemann sum could be defined also with the height of the rectangle being the value of the function for the value of x delimiting the upper limit of the length of the rectangle, Δx. Moreover, (i) a different number of rectangles could be used to span the region delimited by $x \in [a, b]$ (see figure 2(a) vs (b)) and (ii) there is an interdependency on the discrete values of x and Δx chosen within an interval $x \in [a, b]$ (i.e. the IVs in $S(x, \Delta x)$ are not exactly completely independent if used in the frame of figure 2).

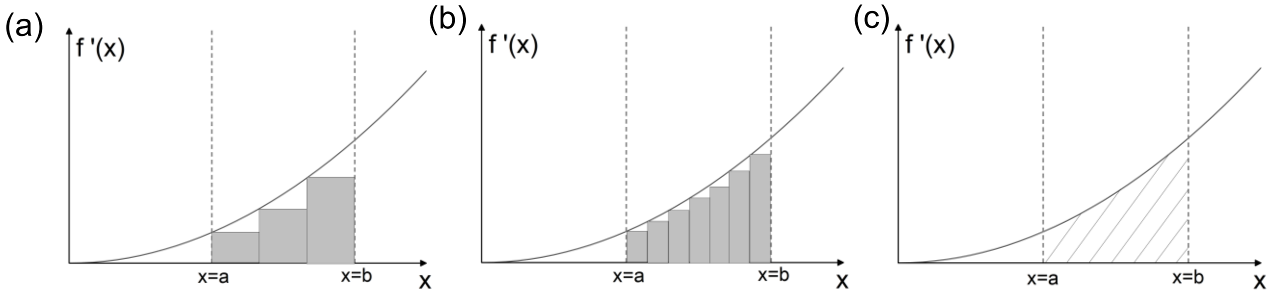

Figure 2. Graphical representation of $f'(x)$ containing vertical lines (dashed lines) to indicate the values $x = a$ and $x = b$ relative to the x axis. Panels (a) and (b) show an estimation of the area underneath the plot of $f'(x)$ within $x \in [a, b]$ made using rectangles. Differences between these two panels are the number of rectangles used and their length Δx. Panel (c) shows the hatched region whose area is trying to be approximated.

$$S(x, \Delta x) = \sum_{i=1}^{n} f'(x_i)\Delta x \qquad (15)$$

It is not difficult to see, upon looking at the sketch shown in the figure, that $S(x, \Delta x)$ is also a way to approximate the area underneath $f'(x)$. This holds for whatever interval $x \in [a, b]$, with it simply being necessary to consider absolute values of $f'(x_i)$ in case it spans negative value so that there is consistency with the concept of area. It is visible that the approximation given by $S(x, \Delta x)$ improves as Δx decreases (meaning more rectangles are used). This is so because the empty space between the rectangles and the function plot diminishes. Notably, this is visibly true, although there is no mathematical expression within basic geometry for the actual area underneath a given general graph and thus no expression for the approximation error to prove in the traditional ways that the error gets smaller as Δx decreases. **It is like reliably visualizing or understanding what happens with the error *without* having an equation that gives the error as a function of the Δx chosen.** So, even in the absence of such an algebraic proof, as long as the conclusions taken from observation or by any other means about the error as a function of Δx are reliable, they can be regarded as solid. In fact, in the context of the latter function, we can conclude by observation (or visualization) that the approximation for the area underneath $f'(x)$ provided by $S(x, \Delta x)$ gets better and better as Δx decreases **without a minimum error threshold**. In other words, the approximation gets **indefinitely** better as $\Delta x \to 0$ (and thus the number of rectangles increases). This overlaps with the concept of limit (value a function approaches indefinitely without ever reaching) to ultimately allow stating equation (16):

$$A = \lim_{\Delta x \to 0} \sum_{i=1}^{n} f'(x_i)\Delta x \tag{16}$$

In words, equation (16) says that the area underneath $f'(x)$ is the value of the limit of $S(x, \Delta x)$ built for $f'(x)$ as shown in equation (16).[2] Notably, this is valid for any interval delimited by $x \in [a, b]$ since, as already stated in the beginning of this discussion, the reasoning made since the beginning of this section is not specific for a given interval of the domain. The only caution is if $f'(x_i)$ spans negative values: in this case, for coherence with the concept of area, it is necessary to take its absolute value.

By calculating the actual limits for different forms of the function $f'(x)$ within a generic interval $x \in [a, b]$, a collection of correspondences between $f'(x)$ and the limit of $S(x, \Delta x)$ when $\Delta x \to 0$ can be obtained. A practical example of finding the form of the limit of $S(x, \Delta x)$ when $\Delta x \to 0$ for $f'(x)$ of a specific form is given below.

Practical example: find the $\lim_{\Delta x \to 0} \sum_{i=1}^{n} f'(x_i)\Delta x$ for $f'(x) = x^2$ within $x \in [a, b]$. We are not going to solve it here, but rather state the answer below ! Please refer to a textbook or to any other material to get a reliable proof – here we are rather interesting on discussing the result.

$$\lim_{\Delta x \to 0} \sum_{i=1}^{n} x_i^2 \Delta x = \frac{b^3 - a^3}{3} \tag{17}$$

In this practical example, we basically obtain in the right hand side (i.e. as a result of the limit we were interested on in the

[2] Note that if we attempted to calculate the value of S(x,Δx) for Δx=0 without any of this context, the function would amount to an infinite sums of zero, which, thinking naively within the underlying definition of multiplication as a mathematical operation, should be zero. This is a vivid example of why ∞.0 is an indeterminate form.

first place) the variation of $f(x) = x^3$ within $x \in [a,b]$. This shows, using this practical example, how it is possible to confirm that the area underneath a function, calculated exactly by the Rieman sum, can alternatively be computed by integrating it over $x \in [a,b]$.

Indeed, determining the correspondence between the function obtained as a result of the limit of $S(x, \Delta x)$ when $\Delta x \to 0$ for functions $f'(x)$ of different forms is an arduous task – as arduous as determining a correspondence between $f(x)$ and $f'(x)$ to establish derivation rules. Yet, once done for several different $f'(x)$ forms, the correspondence can be extrapolated to other $f'(x)$ given by a summation or a product of other functions for which the link with the result of the limit of $S(x, \Delta x)$ when $\Delta x \to 0$ has been made. This is so thanks to properties associated with the limit operation. So ultimately, the arduous task can pay off.

In fact, observing the correspondence between functions $f'(x)$ of different forms and the result of the limit of $S(x, \Delta x)$ built for $f'(x)$ when $\Delta x \to 0$ within a given arbitrary interval $x\epsilon[a, b]$ allows concluding that the result systematically matches the variation of their respective primitive, $f(x)$, within that same interval (i.e. $f(b) - f(a)$). In fact, this equivalence is observed to hold when not treating differently the negative values of $f'(x)$ from the ones that are positive, which is something that we had done when relating the limit case of the Riemann sum to the area underneath $f'(x)$. In other words, evaluating the right hand side of equation (16) in whatever given arbitrary interval $x\epsilon[a, b]$ (even ones contemplating negative values of x$f'(x)$) happens to match the variation of the primitive of $f'(x)$ in that given interval. This means equation (18) holds.

$$\lim_{\Delta x \to 0} \sum_{i=1}^{n} f'(x_i)\, \Delta x = \int_{x_A}^{x_B} f'(x) dx \qquad (18)$$

It is interesting to note that equation (xx) means that the property of having the area underneath $m(x, \Delta x)$ be linked to the variation of the function $f(x)$ it is associated with also holds for the limit case of $m(x, \Delta x)$ as $\Delta x \to 0$. This property had not been initially granted to the limit of $m(x, \Delta x)$ as $\Delta x \to 0$ as it was not possible extrapolate the proof based on the area of rectangles. We also note that the validity of equation (xx) for any function $f'(x)$ was not proved here: **in fact, this is fundamentally not possible as it would be necessary to do it for all function forms there is. Rather, it is argued here that equation (xx) can be proved for different $f'(x)$ forms from the arduous work of linking the result of the expression in the left hand side of equation (16) for a specific $f'(x)$ function to the variation of the primitive of $f'(x)$, namely $f(x)$, within that same interval.** This can be done for the same function forms for which derivation rules were established for consistency.

Finally, the validity of equation (18) as discussed here allows to link the area underneath a function to the definite integral of its primitive. This is because the left hand side of equation (18) itself had already been proved to correspond to the area underneath the graph of any function on which $S(x, \Delta x)$ is built (that is

$f'(x)$ in the case of the left hand side of equation (18)) provided that absolute values of the function are considered. To then ensure that the area underneath a function's plot can be determined via definite integration even when the function features negative values within the interval $x \in [a, b]$ in hand, it suffices to: (i) split the given interval $x \in [a, b]$ so that intervals in which the function is negative and positive are separated, (ii) integrate the function over each and all of the subintervals created in the splitting, (iii) take the absolute value of the definite integrals yielding a negative result and (iv) sum the them with the result of the definite integrals having a positive value. These four steps should be equivalent to splitting the limit case of Riemann sum in the left hand side of equation (xx) into the same intervals and ensuring that in the ones where $f'(x)$ is negative the absolute value is considered. **The four steps thus ensure that the definite integral can be used to compute the area underneath a function.**

b) Angular coefficient of the line tangent to a point of $f(x)$

Within the plot of a given function, it is possible to define a triangle by using two points (x_I, y_I) and (x_F, y_F) alongside a third point defined by the intersection of a fictitious horizontal and a vertical lines. Figure 3(a) illustrates the triangle. Note that the angle is the same regardless if the horizontal and vertical lines pass by (x_I, y_I) and (x_F, y_F), respectively.

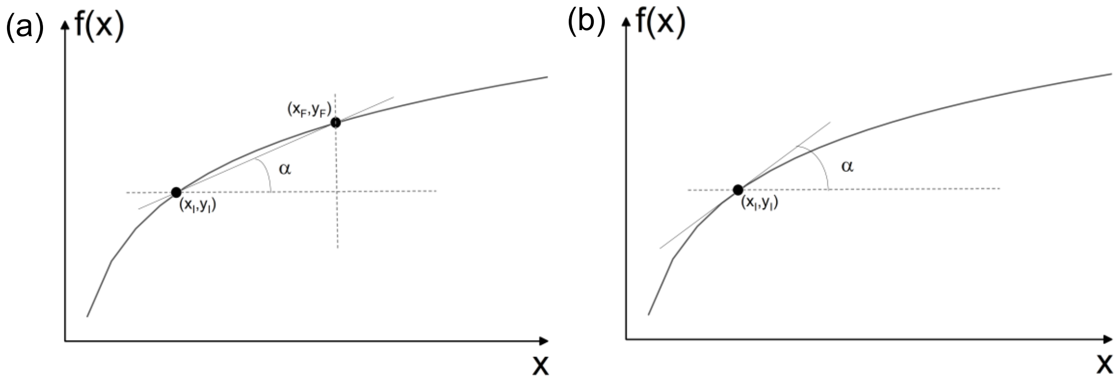

Figure 3. Sketch of the angle α formed by (a) the line that passes through two arbitrary points (x_I, y_I) and (x_F, y_F) which are solutions of $f(x)$ and a line parallel to the x axis and (b) the line tangent to (x_I, y_I) and a line parallel to the x axis.

Within the scope of the trigonometric operations, we have that the tangent of the angle α illustrated in figure 3 can be calculated as shown in equation (19):

$$\frac{y_F - y_I}{x_F - x_I} = \tan\alpha \tag{19}$$

This comes as a consequence of the definition of the algorithm of the tangent of an angle. It is quite straightforward to see by putting together equations (1) and (xx) that the function $m(x, \Delta x)$ also gives the value of the tangent formed in the schematics of figure 3 using any two points of the function $f(x)$ for which it is built. Interestingly, this was not the functionality for which we built $m(x, \Delta x)$ for, but rather a property we notice it also carries (someone, working in a different context than that of average rates, could have crafted $m(x, \Delta x)$ to quantify $\tan\alpha$).

We may proceed to analyse what the limit case of $m(x, \Delta x)$ with $\Delta x \to 0$ quantifies. Again relying in solid observation and visualization rather than a quantitative proof, it is possible to reliably conclude that the value which $m(x, \Delta x)$ approaches without ever reaching matches the angle α formed by a line tangent to the function at a point (x, y) and the vertical and horizontal lines shown in figure 3(b). Again, quantitative proof can be dismissed in face of a reliable proof derived from another source (e.g. visualization).